

Down to Earth

If you live in a rural area in India, you have one of the lowest possible carbon footprints in the world. For a greener life, move out of cities

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Feature

be replaced with CFLs and T5s. Use of Solar Water Heaters can help reduce the amount of energy consumed by geysers. These are few of the steps which can be taken in existing buildings in order to make them green." He adds, "On an average, it takes less than 3 years to recover the extra cost put in the design of an average green home. If designed correctly, this extra cost could be marginal and could be recovered within a few months of completion of the house."

Reiterating Francis' opinion on Indian homes, Vij says, "Indians are slowly going green with their homes. Traditionally, all our homes used to be low-energy, low water consuming and low-waste generating. However, in recent years, due to change in lifestyles, we have shifted away from this traditional knowledge. However, using high efficiency equipment such as BEE star labelled ACs, refrigerators and fans are reducing the amount of energy being consumed in homes. New homes are being designed to minimize water and energy consumption, use renewable energy in the form of Solar Hot Water Heaters and Solar Photovoltaic and adopting practices for rainwater harvesting and recharge. These steps are slowly leading new homes in India to reduce their environmental impact."

The movement

It all started in March 2000 when the when the Government of Andhra Pradesh, invited the Confederation of Indian Industry (CII) team house of Godrej and the Confederation of Indian Industry (CII) came together to start what went on to be India's green revolution among the concrete jungle. The Indian Green Building Council (IGBC) is among the most active green certification bodies in India, certifying for LEED (Leadership in Energy and Environmental Design) in addition to indigenous in-house standards. A humble footprint of 20,000 sq. feet in 2003 has now crossed 365 million sq. feet of green footprint. IGBC conducts certification programmes every month and you can get qualified in the requisite certification standards. Srinivas adds, "you don't need any specialised background in architecture, any person who can read and write can take the course and be certified after qualifying



"Green buildings reduce impacts on site, maximise energy and water efficiency, use sustainable building materials, minimise waste generated and disposed of from the building and provide a healthy living environment for its occupants."

Apoorv Vij

the examination." Quoting an author, he describes a green house as one that gives delight as you enter, serenity and space as you stay and regret as you depart." When around 500 households were surveyed to find out what they expected from a green home, answers varied from energy efficiency, effective lighting to even efficient water utilisation. IGBC's rating systems address these aspects without the need for any sophisticated tools. Talk of practicing what they preach and he gives a first-hand account of his own residence. His home in Bangalore has 13-inch thick walls across the southern and western directions made of flyash bricks that reduce his need for air-conditioning by 50 per cent.

The Government of India and TERI have also jointly developed an indigenous rating system. As Vij explains, "GRIHA is an indigenous green building rating system developed for the Indian construction scenario. It was developed by The Energy and Resources Institute (TERI) and has now been adopted by the Ministry of New and Renewable Energy (MNRE) as the National Green Building Rating System for India. GRIHA incorporates within itself various other building codes and guidelines like the National Building Code, Energy Conservation Building Code, Ministry of Environment and Forests clearance for construction and Pollution Control Board. GRIHA is a rating system which assesses the environmental performance of buildings on a scale of 0-104 points with

a minimum of 50 points required for a building to be certified a GRIHA building. On the basis of number of points scored, a building can be rated between 1 and 5 stars, 1 star being the lowest and 5 star being the highest level of environmental performance. GRIHA evaluates green building performance on the basis of various aspects like water and waste management, energy, site preservation, indoor comfort and air quality and innovation points."

Save all your life

Despite the initial incremental costs, there are incentives from the government that ease out the burden in some ways. The Ministry of New and Renewable Energy (MNRE) has announced various incentives for projects going for



The CII-Godrej Green Business Centre, leading by example

GRIHA rating (explained in the subsequent question). The MNRE refunds the registration fees of projects which achieve a high rating under GRIHA. The MNRE additionally provides cash rewards to the various consultants involved in the design of GRIHA rated buildings which achieve high ratings. The MNRE also provides a subsidy of 30 per cent of the cost of solar photovoltaic modules (up to a 100 kW) which are used in the building. When asked about government incentives, Srinivas had a different take. He added, "of the 500-odd buildings in India, none of them asked for any incentives. Besides, considering that the incremental cost is just 8 per cent, an incentive of 20-30 percent doesn't make a significant difference. Either way the cost is recoverable in a couple of years." 